

Computer Vision - 2026

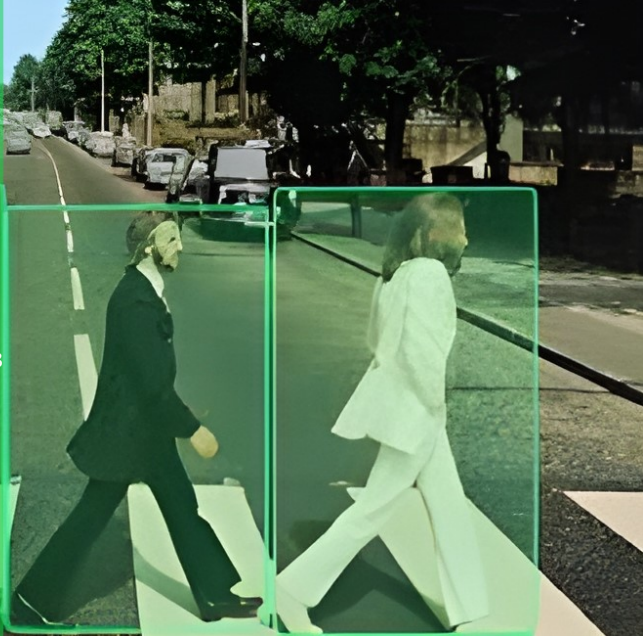
Week #10. Perception of the Dynamic World

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Agenda

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① Representation Learning for Video

② Geometry Models from Images and Video (Depth Anything v3) .

③ 3D Scene Representation -

④ Tracking and Temporal Reasoning .

⑤ From Perception to World Models .

⑥ Unifying Perspective <

Why This Topic Matters

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Classical computer vision treated images, video, depth, tracking, and 3D reconstruction as separate tasks with separate pipelines.

Modern foundation models increasingly blur these borders:

- visual representations can be reused across images, video, and 3D cues;
- depth, segmentation, tracking, and recognition can be built on top of a shared encoder;
- perception is shifting from *frame understanding* to *world understanding*.

Core idea

Images, video, and 3D observations can be viewed as different projections of the same underlying physical world.

From Static Images to Dynamic 3D World Models

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A modern perception stack can be organized as:

observation $\rightarrow$ representation $\rightarrow$ geometry $\rightarrow$ dynamics $\rightarrow$ prediction

- 
- **Representation:** robust visual embeddings from foundation encoders;
 - **Geometry:** depth and multi-view structure from images;
 - **Dynamics:** motion, temporal consistency, and state evolution;
 - **Prediction:** forecasting future observations or actions.

Question for this lecture

Should we still process video frame by frame and 3D slice by slice, or should we learn a persistent world representation instead?

This Week: The Big Picture

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We will study three complementary views of modern perception:

- 1 **Video as temporal representation learning**
how stable embeddings emerge across frames;
- 2 **3D as geometry inference and scene representation**
depth, neural fields, and Gaussian scene models;
- 3 **World models and embodied AI**
how perception supports prediction, planning, and action.

Three recurring questions

- What should be invariant over time?
- What should be explicitly geometric?
- What must be predicted rather than recognized?

Outcomes

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By the end of this week, students will be able to:

- 1 explain why modern video and 3D perception are increasingly built on shared visual representations rather than isolated task-specific pipelines;
- 2 distinguish between frame-wise processing, temporally consistent representation learning, and explicit world modeling;
- 3 describe the roles of foundation visual encoders, depth estimation models, and modern 3D scene representations;
- 4 explain the intuition behind temporal consistency, monocular depth prediction, and continuous scene representations;
- 5 compare classical tracking pipelines with feature-based tracking in embedding space;
- 6 discuss why depth estimation alone is not a full world model, and why dynamics and interaction matter;
- 7 run a modern pretrained model in Colab for video or 3D perception and analyze its strengths and failure modes.

Key takeaway: Modern perception is moving from isolated image tasks toward learning reusable representations of a dynamic 3D world.

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Section 1. Representation Learning for Video

Representation Learning for Video

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Modern video perception is increasingly based on reusable visual representations rather than task-specific pipelines.

Core idea

Video understanding emerges from representations that remain stable across time video and viewpoint.

3d Instead of designing separate modules for detection, tracking, and recognition, a single encoder is trained to produce consistent features that support multiple tasks.

Classical vs Modern Video Processing

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Classical pipeline

frames → detector → tracker → classifier

Modern pipeline

frames → representation → similarity / clustering / prediction

The key shift is architectural: perception becomes representation learning.

Temporal Stability as the Training Signal

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Objects and scenes change gradually between consecutive frames.

$$\underline{f(x_t) \approx f(x_{t+1})}$$

This constraint drives modern self-supervised learning.

A good representation:

- remains stable across time
- changes when object identity changes
- encodes geometry and semantics

Teacher–Student Self-Supervised Learning



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Modern models such as DINO use a teacher–student architecture.

$$x \rightarrow f_{\text{student}}(x) \quad x' \rightarrow f_{\text{teacher}}(x')$$

Training objective:

$$\mathcal{L} = \|\underline{z_s - z_t}\|^2$$

Teacher parameters are updated as an exponential moving average:

$$\theta_T \leftarrow \alpha \theta_T + (1 - \alpha) \theta_S$$

This stabilizes training and encourages consistent representations.

Training Algorithm: Self-Supervised Representation Learning

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Step	Operation
1	Sample frame or image
2	Generate multiple augmented views
3	Compute student features
4	Compute teacher features
5	Normalize feature vectors
6	Compute alignment loss
7	Update student parameters
8	Update teacher parameters (EMA)

Training uses large-scale image and video datasets such as ImageNet, Kinetics, and LVD.

Dense Feature Maps

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Modern encoders produce spatial feature maps rather than a single vector.

$$F \in \mathbb{R}^{H \times W \times D}$$

Each spatial location corresponds to a local visual descriptor.

Objects appear as coherent regions in feature space.

Example: Patch-Based Representation

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A Vision Transformer divides an image into patches.

224×224 image

Patch size:

16×16

$14 \times 14 = 196$ patch embeddings

+ 1

Modern models may compute features at multiple spatial scales.

High-Resolution Dense Features

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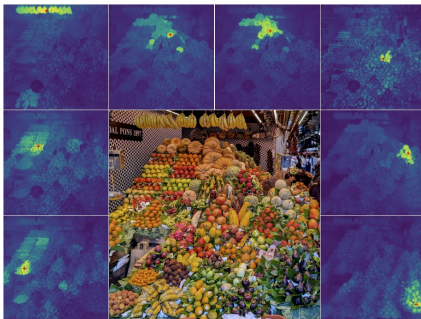


Figure adapted from DINOv3: cosine similarity maps between one selected patch and all other patches. Similarity follows object boundaries and semantic structure.

Interpretation of Dense Features

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Dense representations reveal structure without explicit supervision.

Empirical observations:

- similar regions produce similar features
- object boundaries emerge naturally
- geometry and semantics are encoded jointly

Tracking and segmentation can therefore be implemented as similarity matching.

Tracking as Feature Matching

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Tracking becomes a search problem in feature space.

$$\hat{p}_{t+1} = \arg \max_p \text{sim}(F_t(p_0), F_{t+1}(p))$$

The model identifies the location with the most similar representation.

No explicit motion model is required.

Inference Algorithm: Feature-Based Tracking

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Step	Operation
1	Extract feature map from frame t
2	Extract feature map from frame $t + 1$
3	Select reference region in frame t
4	Compute similarity to candidate regions
5	Select best matching location
6	Update trajectory estimate

$$\text{sim}(a, b) = \frac{a \cdot b}{\|a\| \|b\|}$$


What the Model Learns

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Large-scale self-supervised training produces structured visual representations.

These representations encode:

- object identity
- semantic similarity
- geometric structure
- temporal consistency

The same features can support classification, tracking, segmentation, and depth estimation.

Limitations and Failure Modes

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Representation-based tracking may fail when visual evidence is unreliable.

Common failure cases:

- severe motion blur
- heavy occlusion
- large viewpoint change
- illumination shift

Robust perception requires temporal reasoning and physical constraints.

Paper Reading: DINOv2 and DINOv3

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DINOv2: Learning Robust Visual Features without Supervision

Self-supervised vision transformers trained on large-scale data produce general-purpose visual representations that transfer across tasks.

DINOv3

DINOv3 scales self-supervised vision models and introduces mechanisms that stabilize dense features across resolution and training duration [Oquab et al., 2024; Siméoni et al., 2025].

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Section 2. Geometry Models from Images and Video (Depth Anything v3)

Geometry from Images: Beyond Depth

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Students have already seen depth estimation.

Modern perception systems now infer full geometric structure from multiple observations.

Core idea

Depth is not the goal. The goal is a spatially consistent 3D representation of the scene.

Modern models jointly estimate:

- depth
- camera pose
- scene geometry
- spatial consistency across frames

Conceptual Pipeline: Geometry Model

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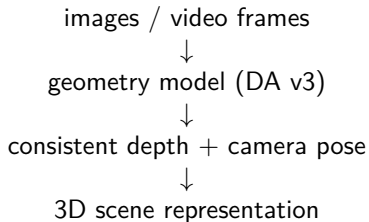
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The model enforces geometric consistency across views.

What Depth Anything v3 Predicts

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Depth Anything v3 is not only a depth predictor [Lin et al., 2025].

It estimates a coherent geometric state of the scene.

Outputs typically include:

- dense depth
- camera parameters
- ray directions
- multi-view consistency

These variables define a 3D reconstruction problem.

Architecture: Geometry Model

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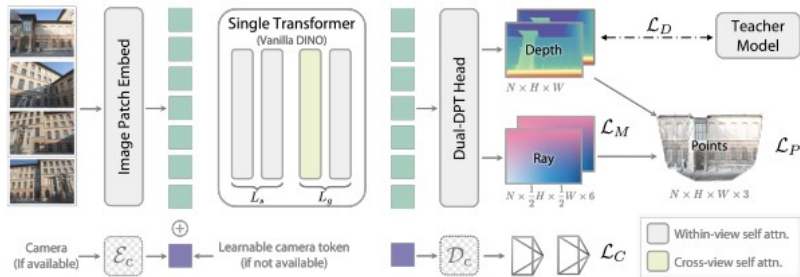
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Architecture adapted from Depth Anything v3.

Core components:

- foundation visual encoder
- geometric reasoning module
- multi-view consistency head

Training Signal: Geometric Consistency

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The model learns geometry by enforcing consistency across views.

Core constraint:

$$\pi_i(X) = x_i$$

where:

- X — 3D point
- π_i — camera projection
- x_i — observed pixel

This is the fundamental equation of multi-view geometry.

Multi-View Geometry Objective

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The model minimizes reprojection error.

$$L = \sum_i \|x_i - \pi_i(X)\|$$

This loss enforces:

- geometric consistency
- correct camera pose
- coherent depth prediction

Training Algorithm: Geometry Model

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Step	Operation
1	Sample image or video sequence
2	Extract visual features
3	Predict depth and camera pose
4	Reconstruct 3D <u>points</u>
5	Compute reprojection error
6	Update model parameters

Inference Pipeline

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Step	Operation
1	Input image or video frames
2	Predict depth and pose
3	Fuse observations
4	Build 3D representation

Inference produces a coherent 3D scene.

Result: Consistent 3D Structure

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image sequence $\rightarrow$ depth + pose $\rightarrow$ 3D geometry

The model estimates the spatial layout of the world.

Limitations

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Geometry inference still has constraints.

Typical failure cases:

- dynamic objects
- reflective surfaces
- extreme motion blur
- insufficient viewpoint diversity

Paper Reading: Depth Anything v3

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Depth Anything v3

Depth Anything v3 learns consistent scene geometry from arbitrary visual inputs and scales to diverse environments without task-specific tuning.

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Section 3. 3D Scene Representation

3D Scene Representation

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After geometry is estimated (e.g., by DA v3), the next step is to represent the scene in a form that can be rendered, queried, or simulated.

Core idea

A scene can be represented as a function that maps 3D location to visual properties.

Modern representations are continuous and differentiable.

Conceptual Pipeline

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images + camera poses



scene representation



render new view

The model learns the structure of the scene rather than storing pixels.

Radiance Field Representation

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A scene is modeled as a function:

$$F(x, y, z, \theta) \rightarrow \text{color, density}$$

Interpretation:

- (x, y, z) — spatial location
- θ — viewing direction
- output — appearance of the scene

This representation enables rendering from new viewpoints [Mildenhall et al., 2020].

Volume Rendering Equation

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Rendered color along a camera ray:

$$C(r) = \int T(t) \sigma(t) c(t) dt$$

Meaning:

- σ — density
- c — color
- T — visibility

Rendering becomes numerical integration along a ray.

Gaussian Splatting: Practical Representation

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Modern systems often use Gaussian primitives instead of neural fields.

A scene is represented as:

- position
- orientation
- size
- color

for each Gaussian.

Why this matters

Gaussian Splatting enables real-time rendering with high visual quality.

Training Algorithm

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Step	Operation
1	Sample camera view
2	Render predicted image
3	Compare to ground truth image
4	Update scene parameters

Training optimizes the scene representation directly.

Inference

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Step	Operation
1	Input camera pose
2	Render rays through scene
3	Generate image

Rendering speed depends on representation efficiency.

Why This Matters

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3D scene representations enable:

- view synthesis
- simulation
- robotics perception
- digital twins

Key takeaway

Depth estimates geometry. Scene representation makes that geometry usable.

Paper Reading: 3D Gaussian Splatting

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3D Gaussian Splatting for Real-Time Radiance Field Rendering

A method that represents scenes as sets of Gaussian primitives optimized for fast rendering and high visual quality.

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Section 4. Tracking and Temporal Reasoning

Tracking and Temporal Reasoning

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Tracking connects perception across time.

Core idea

Tracking is identity preservation under motion and appearance change.

Modern tracking relies on stable representations rather than handcrafted motion rules.

Conceptual Pipeline

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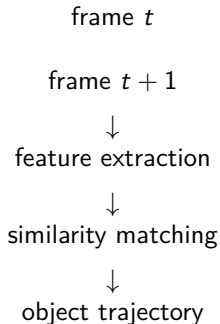
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Tracking is sequential decision-making under uncertainty.

Tracking as Identity Preservation

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The task is to maintain object identity across time.

same object at time $t \rightarrow$ same identity at time $t + 1$

A stable representation enables consistent matching even when:

- position changes
- viewpoint changes
- lighting changes

Similarity-Based Matching

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Object association is performed in feature space.

$$\text{sim}(a, b) = \frac{a \cdot b}{\|a\| \|b\|}$$

Matching rule:

$$\hat{i} = \arg \max_j \text{sim}(f_i^t, f_j^{t+1})$$

The most similar feature corresponds to the same object.

Modern Tracking Pipeline

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video frames $\rightarrow$ detector or segmenter $\rightarrow$ feature encoder $\rightarrow$ data association $\rightarrow$
trajectory update

Tracking integrates perception and temporal reasoning.

Algorithm: Online Multi-Object Tracking

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Step	Operation
1	Detect objects in current frame
2	Extract feature vectors
3	Predict object positions (optional)
4	Compute similarity matrix
5	Solve assignment problem
6	Update trajectories

Assignment is typically solved using Hungarian matching.

Temporal Reasoning

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Tracking requires predicting future state.

Typical state representation:

$$\underline{x_t} = (x, y, v_x, v_y)$$

Prediction step:

$$x_{t+1} = \underline{Ax_t}$$

Motion models reduce ambiguity during occlusion.

Why Classical Tracking Fails

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Failures occur when visual evidence becomes unreliable.

Common causes:

- occlusion
- motion blur
- rapid motion
- viewpoint change

These conditions break similarity assumptions.

Modern Direction: Representation + Memory

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Recent systems combine feature matching with temporal memory.

Examples:

- SAM-based tracking
- transformer trackers
- video foundation models

Memory enables long-term identity preservation.

Paper Reading: SORT

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Simple Online and Realtime Tracking

SORT combines motion prediction and data association to track objects in real time using lightweight computation.

Key insight:

Tracking performance depends more on reliable detection than on complex motion modeling.

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Section 5. From Perception to World Models

From Perception to World Models

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Modern perception systems no longer stop at recognizing objects or estimating geometry.

Core idea

The objective of perception is prediction of future states of the world.

A world model integrates perception, memory, and dynamics into a single predictive system.

Conceptual Pipeline: World Model

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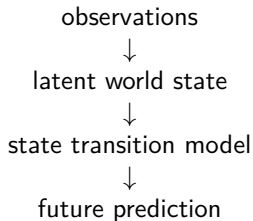
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The system learns internal dynamics rather than static representations.

What Is a World State?

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A world state is a compact representation of the environment.

Typical components:

- object positions
- velocities
- geometry
- interaction context

The state summarizes everything necessary to predict future observations.

State Transition Model

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World models evolve state over time.

$$s_{t+1} = f(s_t, a_t)$$

where:

- s_t — current state
- a_t — action
- f — transition function

This equation defines system dynamics.

Prediction as the Learning Signal

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Training objective:
Predict the future from the past.

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$$L = \|\hat{s}_{t+1} - s_{t+1}\|$$

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Prediction error drives learning of physical structure.

Examples:

- object motion
- collisions
- stability
- trajectory evolution

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Architecture Example: V-JEPA [Assran et al., 2025]

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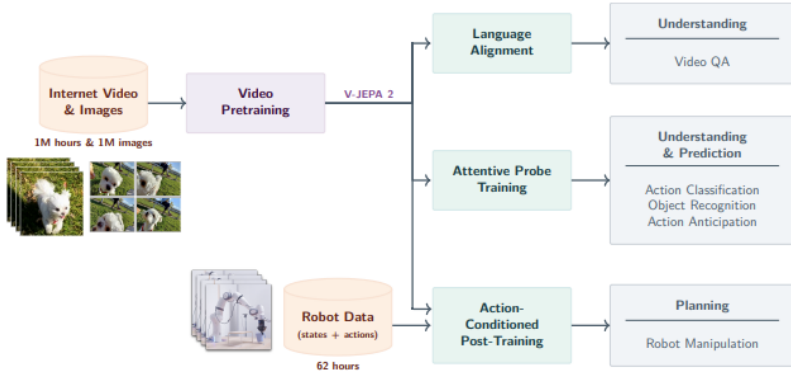
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Architecture adapted from V-JEPA.

The model predicts future visual representations rather than raw pixels.

Key Mechanism: Predict in Latent Space

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Instead of predicting images, the model predicts internal representations.

$$z_{t+1} = f(z_t)$$

Advantages:

- lower dimensional prediction
- improved stability
- faster learning

Latent prediction enables long-term reasoning.

Training Algorithm: World Model

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Step	Operation
1	Observe sequence of frames
2	Encode observations into latent state
3	Predict future latent state
4	Compute prediction loss
5	Update model parameters

Training uses large-scale video data without manual labels.

Inference: Forecasting the Future

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Step	Operation
1	Observe current state
2	Apply transition model
3	Predict next state
4	Repeat for multiple steps

This process produces trajectories rather than single predictions.

How Physics Emerges

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Physical behavior appears when prediction must remain consistent over time.

Emergent properties:

- conservation of motion
- collision dynamics
- object permanence
- stability constraints

These properties are not explicitly programmed.
They arise from predictive consistency.

Why This Matters for Research

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World models enable systems that reason about the future.

Applications:

- robotics planning
- autonomous driving
- simulation
- reinforcement learning
- video understanding

This paradigm connects perception with decision making.

Paper Reading: V-JEPA

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Self-Supervised Learning from Video with Joint Embedding Predictive Architecture [Assran et al., 2025]

V-JEPA learns representations by predicting future visual embeddings rather than reconstructing pixels.

Key contribution:

Prediction replaces supervision.

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Section 6. Unifying Perspective

Unifying Perspective

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Modern perception systems integrate multiple capabilities into a single pipeline.

Core idea

Modern perception is the problem of learning a dynamic 3D world model.

Recognition, depth estimation, tracking, and prediction become components of one system.

Unified Perception Pipeline

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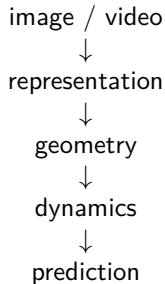
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Each stage builds on the previous one.

The Modern Perception Stack

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Representation → Geometry → Dynamics → Action

Interpretation

Perception is no longer a static task. It is a continuous process of modeling and interacting with the world.

Key Takeaways

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- perception starts from visual representation
- geometry describes spatial structure
- dynamics models change over time
- prediction enables decision making

Final message

The goal of modern computer vision is not recognition. It is understanding and predicting the physical world.

Bibliography

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Mido Assran, Adrien Bardes, David Fan, Quentin Garrido, Russell Howes, Mojtaba, Komeili, Matthew Muckley, Ammar Rizvi, Claire Roberts, Koustuv Sinha, Artem Zhohus, Sergio Arnaud, Abha Gejji, Ada Martin, Francois Robert Hogan, Daniel Dugas, Piotr Bojanowski, Vasil Khalidov, Patrick Labatut, Francisco Massa, Marc Szafraniec, Kapil Krishnakumar, Yong Li, Xiaodong Ma, Sarath Chandar, Franziska Meier, Yann LeCun, Michael Rabbat, and Nicolas Ballas. V-jepa 2: Self-supervised video models enable understanding, prediction and planning, 2025. URL <https://arxiv.org/abs/2506.09985>.

Haotong Lin, Sili Chen, Junhao Liew, Donny Y. Chen, Zhenyu Li, Guang Shi, Jiashi Feng, and Bingyi Kang. Depth anything 3: Recovering the visual space from any views, 2025. URL <https://arxiv.org/abs/2511.10647>.

Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, and Ren Ng. Nerf: Representing scenes as neural radiance fields for view synthesis, 2020. URL <https://arxiv.org/abs/2003.08934>.

Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Hervé Jegou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. Dinov2: Learning robust visual features without supervision, 2024. URL <https://arxiv.org/abs/2304.07193>.

Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, Marc Szafraniec, Seungeun Yi, Michaël Ramamonjisoa, Francisco Massa, Daniel Haziza, Luca Wehrstedt, Jianyuan Wang, Timothée Darcet, Théo Moutakanni, Leonel Sentana, Claire Roberts, Andrea Vedaldi, Jamie Tolan, John Brandt, Camille Couprie, Julien Mairal, Hervé Jegou, Patrick Labatut, and Piotr Bojanowski. Dinov3, 2025. URL <https://arxiv.org/abs/2508.10104>.